Notes on Meeting with LASL + IBM

Console
Simulator
Programming
Manual,
Standard characters

breakpt - pathfinder.

Los Alamos Meeting at Poughkeepsie June 30, '58

(Darwell) Standards - assembled in factory

Transistors ~1/3 Million by end of year { PNP & NPN's

(1) PNP - started first - yields are very good, in plant in Jan.

(2) NPN 6 mos later - in plant in May

main bottleneck now.

~20 days behind.

yield ~ 25% now

expect 40-50% later,

several thousands NPN's, many more PNP's

(10K.)

but circuits use about equal nos.

NPN's work better than PNP's.

(3) - Mem. Drivers: started 3rd. NPN high current.

need relatively few - still in pilot line in Res. - to Plant by end of year.

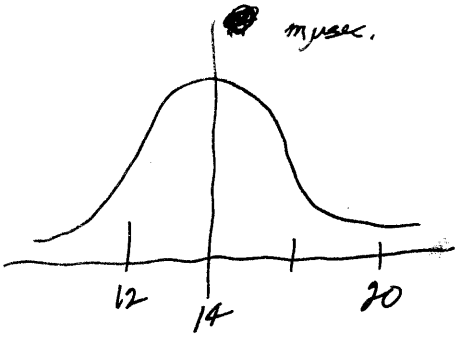
Memory: 2 μs Mem now being fabricated - first model to be completed this year.

- making cores + firing now
- automatic testing
- stringing being done by hand. - x, y machines are being built now.
- several planes available now,

(2.25 μs or faster) — depends on x-stor.

Speed: going 8K — 16K
~ .25 μs slower. still don't know exact values.

Transistors (cont.) performance in circuits, each one has its own peculiarities.



Need full circuit before "sharpening" can occur.

Fast Mem:

Data Flow model later this year
start construction Jan 1959.
not slower than ~0.75 μsec.

Index Core array: — technique not selected as yet.
will not go at 0.2 μs
— time required is obscured by other times.

Transistor mem, still a back-up

(Lazarus :) — cannot contain a code loop.
would like up to 128 —
.. complete etc

Transistor: PNP, NPN's

no diodes, no condensors, — very primative,
no level setting required,
later cost of Transistors should approach cost of good diodes. (diodes are not as high speed.)
also are point contact devices

basic circuits. 12
pluggable units. ~24
} basic catalog.

also 2 new classes (1) 3 level logic — cascode, more in one stage faster.
(2) emitter follower

- bench testing has been going on. testing approval in past few weeks.
- circuit cards produced in plant,

approval — end of life — marginal testing
— noise coupling on card

(Harvard) cap - cards.
1 - 2 way "and's" shown

two types NPN, PNP.
about 90% of machine can be made of four of these.

no trigger as such. Two coupled back to back.
most Trigger are 3 way ands,
and or and
← latch back

- Studies on larger card going ahead.
for adder stage — registers etc } as yet unapproved

Frames: Exchange: laminar bus,
panels wired at plant in week or two,
- some detail wiring

8 channel exchange - will be ready as soon as have enough X-ISTORs

Memory: - first fire-up in few days. - have enough X-istors.
(Tour of annex) - ground planes.
- noise.

~ 1 Oct. exchange should be running. - will dark large sections of machine.

no emitter-followers in Exchange.
Schedule for rest of machine.
Data flow model of arith. system. 6 bits, & part of controls
(cut thru adder & multiplier.),
will be done with standard pluggable units first
may use special cards later.
- Wired starting this week - will be ready before transistors are available.

Information before Oct 1.

(Bloch) Detail design: Mem Bus Start 6-30 Power on 9-30
Schedule. Indexing Unit (incl look-ahead) 8-1, '58
Arith Unit: 1-1-'59

arithmetic Speeds:

we will talk about conservative speeds
— truth will be inbetween these & contract speeds.
certain speeds now are:
Flpt add 0.95

Estimates of No. Transistors. = 140,000,
(not Bus, Mem, Exchange)

Floating Point Arith	25%	mostly data
VFL Arith unit	15%	mostly control
I-box	17%	
Look Ahead	17%	mostly control,
Checking	26%	

Register positions ~1000, no register sharing.
adder positions ~ 310,

I Box: 0.4 μsec basic cycle.

accessing instructions, also has two buffer
modifys instrs, registers A, B
executes index with instrs for I Box
modifies memory addresses (box, no) F.U.-MM
compares address (range)
all memory references (data) — return to LA
Transfering updated instrs to Look Ahead.

→ Rate 0.8 μs per instr. for floating pt.

Index Core a 2 Dimensional Mem.

two alternatives (a) a destructive read 0.4 – 0.4 read out regen.
(b) non-destructive 0.2 – 0.6 read only writing (destroy then write)

Look-Ahead:

4 levels.

VFL takes up 2 levels

VFL crossing wd. boundaries – 3 levels.

- stores all instr to be granted by AU
- fetches operands.
- performs all store operations (data from AU) (except index core storage)
- ~~Cache~~ contains as pseudo stores.

quantities stored in Look Ahead.

- I.C.
- unmodified Data Field (Index word V+C unchanged)
- Operand address
- Updated indicator field
- Op. code
- operand
- (tags OF bit, checking bit.

L.A. cycle time = 0.2 μs level to level, level to AU.

Arithmetic Unit:

— only connection to outside world is Look Ahead.

AU is asynchronous w.r.t. LA.

(a) VFL loop → (cycle less than 0.45 μsec)

1. executes all VFL binary ops, except mpy & div
2. executes all VFL decimal ops,
3. executes all Connect ops,
4. executes exponent arith for FL Point.

[binary parallel 8 bits
decimal 1 digit serial]

Times:	8 bits	32 bits
Binary add	1.35 μs	2.7 μs
Binary Mpy	9. μs (prod 16)	12. μs (prod 64)
Binary Div	12. μs (div 16)	19.8 μs (div 64)

	5 digits
Decimal add	3. μs
Decimal Mpy	63.
Decimal Div	87.

question:

VFL LD
VFL MPY 10.35

VFL LD 1.35 μs
FL STO .5
VFL LD 1.35
VFL Mpy 1.7
4.9

(Square) may have to use VFL PT for integer work — due to speed.

Arith. Unit: (cont)

(b) Parallel

performing FP mantissa ops.

" VFL Mpy + Div (binary)

(consists of a shifter: 0, ±1, ±2, ±3, ±4 (doubtful ±5, ±6) now +12

adder: 96 bit parallel

Mplier: 12 bits of Mplier in 1 cycle (4 cycles for 48 bit using same adder. fixed time.

average Fl. Point add (once thru shifter only) — pre + post shifts. (2 cycles total)

Normalized add: 0.95 μs

shift cycle ~0.2 μs

unnormalized add (~0.2 μs less)

register
↓
shifter
↓
adder
↓
register.

(Syzygy) question: if there is a pre-shift + no overflow. There may be one a lead zero. can we leave out extra shift?

(Block) — controls would be harder --

Discussion: on pre + post shifts

question: do larger pre-shifts become more imp. on larger probs. — smaller integration

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Fl. Normalized Mpy	1.75 μs	all ind. clocking.
Fl. Divide	7.0 μs	includes one time thru shifter.
Fl load-store	0.35 μs	

Checking: data Mem : Hamming,

registers + registers : parity

VFL adder, duplication

FL. adder, multiplier } casting out 3's process,

→ checking adds 1 cycle on each :

costs about ~~.2~~ .150 μs per operation

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note: → no provisions to repeat automatically

~~as~~ so as not to hold up all ops, when there is no

— asynchronous technique : can

IBox ~~sync~~ Mem. Bus ↔ Mems

AU ↔ LA 0.2 synchronized.

IBox 0.4 cycle, Bus 0.2 μs cycle locked together

reclocking is avoided.

Kelsey — talk on Simulator.

(Dunwell) Cryogenics — was examined ~ Jan

— but Technical prog. — indicates

lasers 16 Thick — power losses at high rep. rates.

— probably will have cryogenic memories fast —

Stretch — Memories could be substituted:

Discussion of Manual:

- I/O: Disk times ?

(Sweeney)

☐ Reg. channels — 8 per module — 9 bits wide

expandible to 32 (by 8's)

"Type B" → (another possible @ 17 or 32)

Low Speed channels — up to 64 units

High Speed Exchange — 2 channels read-write (not read-read write-write)

8 μs per wd
1 Million wds. } for sure

may be able to go to double track density 2M

- less chance of going to double bit 4M at 4 μs

cannot have simultaneous read/write

→ Disk Del. Date : ~~Oct~~ Oct '59 won't hear for some time

Card Reader : 1000 cpm — incl. adapters, transitions, & buffers.

Card Punch : 250

Tape adapters : 2 adapters 2 tapes each

729 I at least.

- dual density tape possible. 729 I, 729 III.

console & IQS :

Printers : 1000 lpm printer, & 500 lpm pr. not too reliable.

were using 150 lpm as temporary situation.

seems to have 600 lpm coming.

132 char per line at 600 line per min

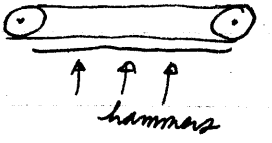
expanded char set. slows printer down.

Punching Typewriter, also IRS } 88 characters. about 20% reduction.

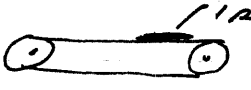
but IRS doesn't need Hollerith.

question: how important are other characters on 600 lpm printer.

chain: 5 sets of 48



Special characters



600 lpm.

480 lpm. — cost questions — we don't know

150 lpm if all special characters.

questions of CRT printer?

Stromberg printer?

(Bengt) want standard printer • 48

inquiry station -

(present 88 RPT can be considered only as temporary) set.

lower case alphabet is important

120 characters + up.

Needs:

(1) → very large vocab, at low speed (report writer)

(2) → 48 char vocab high speed (table writers)

(Bengt) suggest full 240 choices. (speed not important)

automatic ext-counter eliminated. — very well received,

address monitoring : exchanges can write — address not compared

Supervisor : —

compiler monitoring ----

question of "inconstant" results - eg. progressive indexing with same I,

→ Breakpoint bits: extra bits

now. page 2,20 "modulo 2^{16}" instead of "2^{18}"?
can this be changed in machine —?
(probably not)

(Sozpuer) — being able to tag instrs. is more important
than having 2 bits [redacted] unused.
— may have to exclud inadvertes

addrs compare — not considerd important

Rename —

question of rename X13 — then late X11 — get interrupt?
— possible errors? not too important.

page 2.35 : return to same instr – break-in "no ops" in interrupt table.
can get into loop. – (time clock breaks-in.)

2nd Day Tues July 1.

Console: 64 binary toggles
64 " lites
left & right address banks.
reset button.
clock –

Suggestions:

1. slide keys back (retractable) – let flow design slide under.
2. books going over switches.
3. Discussion of number displays.
4. question of Hexadecimal.
5. back of Typewriter should be raised – to read current line.
6. display internal clock in address field
7. Pseudo pots. – be sunk into desk with plastic cover – retractable?
8. Sliding arms
9. clock can be higher.
10. "Stop" should not be

Initial Load Interrupt:

should be fixed. ⇒ destroys instr. counter — why can't it hold until STICBR is done.
(only machine checks are higher than I/O interrupt)

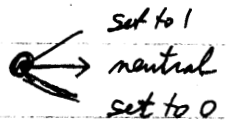
- Force STIC-BR into zero.
- Then give a "read" into one.

— "stop" key, console is unready.

(bazarre) Sense switches: ref. contract "large no. — some of which are" — [a dozen or so are enough]

- read by machine a operate on per basis
set but not held, to 1 or 0

a three pos. switch — needed. — 2 bits to represent in storage.



going three exchange is OK. ←

or need a flip-flop — relays?

Keyplex — use for "arguing" with machine countermanding.

—

→ line to read in word directly — external units.

also need

lights dependent on switches ← (this is what they want)
separate from dependent

illustrated display — not so important — can use typewriter.
The "onpon" switches are important

— an interpretive console. OK.

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programing Session:

- assembly - Walton, Sandgath
- auto.prog - Voorhees, Wells "ALGAE I" — pre-processor. 32K no drums.
- supervisor - Heising, apple

In August - "universal language" report.

- assumes no spacial placement - blank isn't meaningful.

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assembler — start simple on 704

— grow with time put on STRETCH

q: is emphasis on hand prog.? or automatic prog

a: start on 1 to 1 hand prog. - OK on 704

later on macro-ops - fatten. - needs STRETCH

(Voorhees) how much "middle ground" programs?

STRAP coder: questions.

(1) variable length vs 6 character limits (should not restrict fields)

(2) coding form

C	LOC	Basic OP	MOD (all lumped)	ADDR	I, J, K (second I)	2nd Addr (L,B,O)	Fm	comments
	HYDRO							
relative →	HI							
absolute →	75241							

blanks

addressing:
Regional
Relative
Compound

fixed field — may be OK for

Example:

	LOC	op	Mod	Addr	comments
		L		R,I [2]	GEOMETRY CALC
		A		R-1,I	
		DIV		L(2)	
		S		RBAR,I	
		LX		DATA,,J	

Suggestion: J is really an "adverb"

op	J	Addr I

"load J from A,I"

→ one can write more than one op or statement on a line — variable

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(1) Field Code vs. Variable fields
(2) Field Separators.

→ ability to stack programs — loaders & Tr cards an important help.

Decisions on vertical lines — does assembler assume vertical line?

1. Extent of Columnization
   - 1.1 No columns for guidance only
   - 1.2. Loc, op, addr, comments
     - For guidance only
     - cols set by programer
     - Fixed
   - 1.3 completely columnized
     - for guidance only
     - set by Pragma
     - fixed
2. Field Separators.
   - 2.1 one kind — , / — 2.11 LINE or COMMA ← ?
   - 2.2 two kind
     - blanks, comma
     - lines, comma
     - other, (letter, Typewriter)

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use of space

append on (1) col 1 special
(2) = overwritable blank for
- other lines ?
  comment field ?

→ Will Try some examples, coding

Addressing:

- 1st char. is a letter - others alphanum
- locs only at start of region

partial reassembly:

(Walton) case 1. assign INST LOC from Loc counter
case 2. " LOC CTR from INSTR CTR.
and saying "LOC EQU NEXT"
HYDRO · EQU 5000.
go to table. } Partial Reassembly

Case 3. assign loc. CTR arbitrarily
(1) Absolute LOC EQU 5000
(2) Symbolic LOC EQU Symbol
(3) Relative LOC EQU LOC+100
(4) Compound LOC EQU HYDRO+(A2-A1)
(5) definition of long symbols.
Substitution (Ditromaks)

Need to make decisions on

1. Coding form
2. Card Form
3. Mnemonics
4. output form
5. pseudo ops (e.g. BSS, ORG, DEC, LIB, etc.)

George Grosz & Walton will discuss & try to formulate system.

on supervisor program :

1st try : one program + using peripheral equip separately. cd. to tape

final ; multiprogramming

supervisor — console defining program.

interaction between — supervisor & assembly.